

Product Requirements Document

AI Call Agent System

College Administration Outbound Calling Platform

Document ID:	PRD-AICA-001
Version:	1.0
Date:	October 2026
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Status:	Final
Stakeholders:	College Admin, IT Dept, Faculty

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1 Executive Summary

1.1 Problem Statement

College administrative staff spend significant time making routine phone calls to students and parents for fee reminders, exam notifications, attendance alerts, and event announcements. These calls are repetitive, time-consuming, and inconsistent in delivery. Staff availability limits the number of calls that can be made daily.

1.2 Solution

The AI Call Agent is an on-premises automated telephony system that places outbound calls on behalf of college administration. The system conducts natural spoken conversations in English, Hindi, and Gujarati using local AI models, delivers consistent messaging, and scales to hundreds of calls per day without additional staff time.

1.3 Success Metrics

hdr	hdr	hdr
Calls placed per day	20–30 (manual)	100–200 (automated)
Staff time per 100 calls	4–5 hours	15 minutes (scheduling only)
Call success rate	N/A	≥ 85% answered
Cost per call	Rs. 5–10 (staff time)	Rs. 0.76 (Plivo)
Languages supported	1–2	3 (En/Hi/Gu)
Monthly operating cost	Rs. 0 (staff time hidden)	Rs. 500–800

2 Product Vision

2.1 Vision Statement

To become the college's primary automated communication channel — an AI voice assistant that speaks to students and parents like a knowledgeable staff member, in their preferred language, at any scale, at near-zero marginal cost.

2.2 Target Users

Primary Users (Staff):

- Administrative clerks scheduling fee-reminder calls
- Department coordinators sending exam notifications
- Placement cell sending interview schedules

Secondary Users (Recipients):

- Students (B.Tech, M.Tech, MBA)
- Parents/guardians

2.3 User Journey

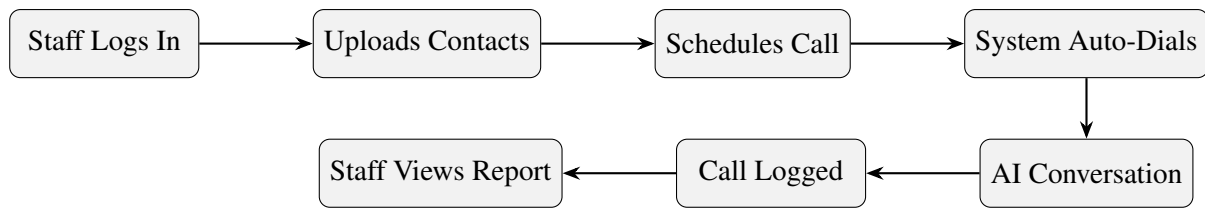


Figure 1: Staff user journey — end to end

3 Features and Epics

3.1 Epic 1: Core Telephony Pipeline

Goal: Place real PSTN calls and conduct AI conversations.

hdr	hdr	hdr	hdr	hdr
US-01	As a system, I place a call via Plivo at scheduled time	P0	M	1
US-02	As a system, I bridge answered call into LiveKit	P0	M	1
US-03	As a system, I transcribe speech with IndicConformer	P0	M	1
US-04	As a system, I generate reply with Qwen3-14B	P0	L	1
US-05	As a system, I synthesize speech with IndicF5	P0	M	1
US-06	As a system, I play audio back to caller via Plivo	P0	M	1
US-07	As a system, I end call gracefully on completion	P0	S	1

Effort: S = Small (1–2 days), M = Medium (3–5 days), L = Large (1–2 weeks)

3.2 Epic 2: Staff Web Interface

Goal: Allow non-technical staff to schedule and monitor calls.

3.3 Epic 3: AI Model Management

Goal: Ensure AI models run reliably with fallback to cloud.

3.4 Epic 4: Call Scripts and Templates

Goal: Allow staff to define what the AI says in different contexts.

hdr	hdr	hdr	hdr	hdr
US-08	As staff, I log in with my username and password	P0	S	1
US-09	As staff, I add/edit/delete contacts	P0	S	1
US-10	As staff, I upload contacts via CSV	P1	S	1
US-11	As staff, I schedule a call with date, time, contact, purpose	P0	M	1
US-12	As staff, I cancel a scheduled call	P0	S	1
US-13	As staff, I see live status of all active calls	P1	M	1
US-14	As staff, I view call history with transcript	P1	M	2
US-15	As staff, I export call report as CSV	P2	S	2
hdr	hdr	hdr	hdr	hdr
US-16	As a system, I fall back to Groq if LLM exceeds 5s	P1	S	1
US-17	As a system, I fall back to Sarvam AI if STT fails	P1	S	1
US-18	As IT admin, I see health status of all AI models	P1	S	2
US-19	As IT admin, I update Ollama model without downtime	P2	M	2
hdr	hdr	hdr	hdr	hdr
US-20	As staff, I select a call purpose (fee reminder, exam notice, etc.)	P0	S	1
US-21	As admin, I create and edit call script templates	P1	M	2
US-22	As admin, I preview how AI will handle a script	P2	M	2

4 Call Script Templates (Phase 1)

The following predefined call purposes are included in Phase 1:

hdr	hdr
Fee Reminder	Informs student of pending fee amount and due date; asks for confirmation; notes response
Exam Notification	Informs of upcoming exam schedule; answers questions about date/time/venue
Attendance Alert	Notifies student/parent of low attendance; explains consequence; asks for reason
Event Announcement	Announces college event; provides date and venue; takes RSVP
Placement Notice	Informs of placement drive; shares company name and date; confirms attendance
General Inquiry	Open-ended template for custom messages; staff defines opening message

5 Release Plan

5.1 Phase 1 — Core MVP (Weeks 1–6)

- Full AI telephony pipeline (STT + LLM + TTS)
- Plivo integration for outbound PSTN calls
- Basic web UI: login, contact management, call scheduling
- Call logging to PostgreSQL
- Audio storage to MinIO
- Docker Compose deployment
- 6 predefined call templates

5.2 Phase 2 — Enhancements (Weeks 7–12)

- Dashboard with charts and analytics
- Bulk call campaigns with contact groups
- Call transcript viewer in UI
- CSV export of call reports
- Custom script template editor
- Email alerts on call failures

5.3 Phase 3 — Future (Post-deployment)

- WhatsApp/SMS follow-up after unanswered calls
- ERP/LMS integration (auto-import student lists)
- Multi-DID support for parallel calls
- Call sentiment analysis
- Voice clone for personalized AI voice

6 Cost and Pricing

6.1 One-Time Costs

hdr	hdr	hdr
Domain name (1 year)	Rs. 800–1,200	.in or .com
SSL Certificate	Rs. 0	Let's Encrypt (free)
Plivo DID setup fee	Rs. 0–300	One-time activation
Server hardware	Rs. 0	Use existing college PC
Total one-time	Rs. 800–1,500	

6.2 Monthly Recurring Costs

hdr	hdr	hdr
Plivo DID number rental	Rs. 200	1 India number
Plivo outbound calls	Rs. 0.38/min	Pay-as-you-go
Groq API (LLM fallback)	Rs. 0	Free tier sufficient
Sarvam AI (STT fallback)	Rs. 0	Free tier sufficient
Server electricity	Rs. 200–400	GPU at 60W avg
500 calls × 2 min	Rs. 780	Rs. 380 calls + Rs. 200 DID + Rs. 200 power

7 Risks and Mitigations

hdr	hdr	hdr	hdr	hdr
R1	GPU hardware failure	Low	High	Keep cloud fallback (Groq) always configured
R2	Plivo API outage	Low	High	Monitor; alert staff; retry queue in Celery
R3	Model accuracy poor in Gujarati	Medium	Medium	Test with 50 native speakers; fine-tune if needed
R4	Students reject automated calls	Medium	Medium	Identify caller as AI in opening; allow opt-out
R5	TRAI regulatory issues	Low	High	Disclose AI caller; maintain DND compliance list
R6	Internet outage at college	Low	Medium	Queue calls; retry when connection restored

8 Acceptance Criteria

The system is considered production-ready when all of the following are met:

1. A staff member with no technical knowledge can schedule a call in under 2 minutes
2. 10 consecutive test calls complete successfully end-to-end
3. AI responds correctly in all three languages (En/Hi/Gu) on test scripts
4. End-to-end latency per turn is below 5 seconds on college hardware
5. Call log is visible in UI within 10 seconds of call completion
6. System recovers automatically after a simulated crash within 60 seconds
7. No plaintext credentials found in codebase (security audit)